- Confidential -

Simpo Full-Stack Engineer Coding Exercise

Welcome to the Simpo coding exercise. The purpose of this exercise is to demonstrate how you write code in real life, outside of a whiteboard interview.

Please treat it as an opportunity to showcase your strengths. If you lean much more towards front-end development, please dedicate special attention to the front-end UI and experience. If you're stronger on the backend, dedicate more time to that part.

Additionally, given it is an exercise and not a production system, you don't need 100% unit test coverage, solutions to all edge cases, or even feature-completeness (though the more you do, the better).

Overview

You will implement a simple web application to analyze earthquake data.

You will use a 3rd party API to fetch information about historical worldwide earthquake data, which accepts a variety of parameters, as follows.

The API

Here's an example of an API GET request:

https://earthquake.usgs.gov/fdsnws/event/1/query?format=geojson&starttime=2000-01-01&endtime=2020-01-02&minmagnitude=5&minmagnitude=1&latitude=37&longitude=100&maxradiuskm=200

Explanation:

format – geojson is a standard json representation (https://earthquake.usgs.gov/earthquakes/feed/v1.0/geojson.php)

starttime – will return earthquakes happened after this date YYYY-MM-dd

endtime – will return earthquakes happened before this date YYYY-MM-dd

minmagnitude – will return earthquakes with magnitude larger than the param

location – will return the earthquakes happened in a circle with the radius in km and centered in the given coordinates.

latitude – [-90,90]

longitude – [-180,180]

maxradiuskm – radius in KM

Try this request out in the browser to see the structure and values for yourself.

you can read about the service API in this link:
https://earthquake.usgs.gov/fdsnws/event/1/

Scope

For this assignment, we will not be sending requests directly from the client side to the 3rd party API. Instead, we'll send the requests to our own custom backend server, which will essentially act as a proxy for the client to communicate with, and relay the requests to and from the external API.

For the client-side, create an interface to accept the following parameters from the user:

1. Start date
2. End Date
3. Minimum magnitude – decimal 0-10
4. Location
    4.1 Latitude – Decimal [-90,90]
    4.2 Longitude – Decimal [-180,180] 4.3 Radius – Decimal (km)

Based on the submitted data from the client, you should display information about all earthquakes that happened:

- In the chosen date range

- With greater magnitude then the minimum magnitude selected
- In the circle created by the location and radius

Once the data is retrieved, display the:

- Median magnitude of all earthquakes
- Max magnitude
- Min magnitude
- Number of earthquakes
- A table of the first 100 results with columns of: place | magnitude | time
  (you can add pagination, or display all of the data at once)

In the exercise, you should strive for:

- Readable code
- Proper error handling
- A separate front end application and backend server
- Efficiency

A quick reminder:

Play to your strengths. Are you unparalleled in your backend skills? Go for a minimal frontend. Are you a phenomenal frontend developer? Put the bulk of your logic in the frontend.

We're rooting for you!

Good luck,
Team Simpo